CS F364: Design & Analysis of Algorithms

Quiz 2 – June 4, 2026

Coverage: Lectures 1–10

> Note
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> **Instructions:**
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> - Time: 15 minutes.
> - Write your answers clearly on paper, scan, and upload to Google Classroom.
> - Show all steps for full credit.

Question

Consider a modification of standard Merge Sort, called **3-way Merge Sort**, which works as follows: - **Divide:** Divide the input array $A[1..n]$ into three subarrays of size roughly $n/3$. - **Conquer:** Recursively sort each of the three subarrays. - **Combine:** Merge the three sorted subarrays into a single sorted array.

(a) [1 Mark] How many comparisons are required in the worst case to merge three sorted lists of combined size n into a single sorted list? Explain briefly.(Use exact values, including constants)

(b) [1 Marks] Write the recurrence relation for the worst-case running time $T(n)$ of 3-way Merge Sort and state its overall asymptotic complexity in Θ notation.

(c) [3 Marks] Compare the running time of 3-way Merge Sort to standard 2-way Merge Sort: 1. What is the exact running time in worst case of 2-way merge sort. 2. What is the exact running time of 3-way merge sort. 3. Which version is more comparison-efficient? (Hint: $\log_3 n = \frac{\log_2 n}{\log_2 3} \approx 0.63 \log_2 n$).